Equivariant Self-Supervised Learning for Dynamic 2D MRI Reconstruction

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Abstract

Reconstructing dynamic MRI sequences is an intrinsically challenging task as the nature of the acquisition prevents us to obtain high quality measurements due to the live capture of Fourier coefficients that can't cover the entire frequency spectrum. This is why these reconstructions are of poor quality with strong motion artifacts, in opposition to anatomical MRI where we have more time to swipe a wider range of Fourier coefficients for a static volume acquisition. State-of-the-art methods still imply supervised learning but suppose we possess high quality measurements. Equivariant Self-Supervised Learning (ESSL) models have demonstrated the possibility to extract prior knowledge from raw measurements only and obtain similar results as supervised models in diverse inverse problem scenarios such as image inpainting, CT, MRI... but previous works have always been applied to 2 dimensions problems. We extend Equivariant Imaging to Dynamic MRI, introducing a novel temporal transformation to enhance reconstruction in spatio-temporal domain. Our Dynamic ESSL (DESSL) models, based on an unrolled Plug-and-Play (PnP) - ADMM architecture with DnCNNs, are tested on the QTIM dataset with simulated MRI degradation. Using PSNR as a metric, we demonstrate that DESSL approaches the performances of fully-supervised methods.

1. Introduction

Equivariant Self-supervised learning [1] represents a cutting-edge approach to address reconstruction inverse problems [2], which are characterized by the need to recover high-quality images x from under-sampled and noisy data y, inverting the process:

$$y = Ax + \epsilon \tag{1}$$

with A being the forward operator and ϵ some additional noise. The challenging part of this task lies in the ill-conditioned forward operator A, meaning that there are potentially infinitely many images x that could have produced the observed data y. The objective is therefore to retrieve information about the null-space of A denoted $\mathcal{N}_A$ to limit the range of potential solutions.

Historically, model-based approaches have been proposed to converge toward a convenient solution through fixed regularization techniques and proximal algorithms. These methods rely on strong assumptions regarding the desired solution. However, with the emergence of deep learning, a more modern approach is to employ deep learning in order to learn the whole reconstruction. With the introduction of learned priors (e.g. PnP) integrating a neural network for denoising (e.g. DnCNN [3]), we can learn a denoiser from data only. In order to learn every parameters of the reconstruction algorithm, including the ones of the prior, it is now common to adopt an unrolling optimisation procedure [4] in an attempt to learn a general denoiser. These learned models are in general based on pairs of ill-conditioned operators and ground truth operators to get information about A in a supervised framework. However, solving an inverse problem using ground-truth data is ambiguous as in principle, in many real world problems, we do not possess this ground-truth data (especially in the case of dynamic MRI). This fundamental issue is tackled in [1], enhancing the challenging task of learning the reconstruction only thanks to noisy measurements, and introducing the potential of Equivariant Imaging (EI) to learn beyond the range space.

EI leverages the concept of equivariance, which refers to the transformation properties of data under specific geometric operations, such as rotations, flips and projective transformations [2, 5]. By exploiting equivariance, EI frameworks can effectively model the underlying physics of the imaging process and incorporate prior knowledge within the measurement data. EI can be achieved through the architecture of a learning model or directly through the learning process, and showed promising results. Learning with EI in a self-supervised architecture means we

exploit equivariance to obtain information about the null-space of A, suggesting we learn beyond the range of A^T [1].

The particular nature of MRI measurements, acquired in the Fourier domain before being transformed in the image domain, adds difficulties to handle the data (for patching with small computing power) but previous experiments over the 2D knee MRI dataset (fast MRI) with robust equivariant imaging [6] demonstrated more than convincing results in presence of empirically assumed Gaussian noise.

In this research, we study the behavior of EI and its variants applied to dynamic MRI data with state-of-the-art Unrolled PnP-ADMM architectures, involving the deployment of a new group transformation with affine spatial transformations combined with a temporal flip transformation. Also, we develop smart patching methods in the specific case of dynamic MRI to deal with memory-consuming unrolled models on GPU.

2. Related Works

2.1. Equivariant Imaging Principle

Learning a solver h_θ in a self-supervised fashion is impossible without using strong assumptions on the set of possible solutions [7]. Therefore, if we dont have access to the ground-truth measurements in a framework without invariance, there is no information we can extract to learn the null-space. However, in the context of EI, each measurement can be seen as a new observation with a new measurement operator $A_g = AT_g$:

$$y = Ax = AT_g T_g^T x \tag{2}$$

Using $\tilde{x} = T_g^T x$, also a signal in $\mathcal{X}$, yields $y = A_g \tilde{x}$ and the invariance property allows us to see in the range of the operators A_g and not only in $\mathcal{R}_A$. This can be understood as the invariance rotates $\mathcal{R}_A$ under the action of $\mathcal{G} = \{g_1, g_2, \ldots, g_{|\mathcal{G}|}\}$ such as:

$$\mathcal{R}_{A_g} = range\left(T_g^T A^T\right) = T_g^T \mathcal{R}_A \tag{3}$$

meaning we now see in a bigger range space $\cup_g \mathcal{R}_{A_g} \supseteq \mathcal{R}_A$. Ideally, we would like to reduce the null-space to a trivial one. Therefore, we can deduce the necessary condition to recover $\mathcal{X}$ from compressed measurements, which is that the matrix:

$$M = \begin{bmatrix} AT_1 \\ \vdots \\ AT_{|\mathcal{G}|} \end{bmatrix} \in \mathbb{R}^{|\mathcal{G}|m \times n} \tag{4}$$

is of rank n. This yields the lower bound on how large $\mathcal{G}$ has to be with $m\,|\mathcal{G}| \geq n$. We need to be vigilant as not any combination of forward operator and group action verify this condition [7]. A necessary condition is that $\mathcal{R}_A$ cant be invariant to all $g \in \mathcal{G}$ (which is required to learn beyond $\mathcal{R}_A$), therefore there is one g that satisfies: $\mathcal{R}_{A_g} \neq \mathcal{R}_A$.

2.2. Equivariant Imaging Loss

In [1], authors propose the vanilla equivariant imaging loss, but developed different variations in [6], with the EI_1, EI_2, and REI losses. Exploring the original EI loss, being composed of the data consistency part and the term to impose equivariance, we try to minimize:

$$\mathcal{L}_{EI}(\theta) = \mathcal{L}_{MC}(\theta) + \lambda \mathcal{L}_{EQ}(\theta) \tag{5}$$

Where the measurement consistency is denoted by the error function [6]:

$$\mathcal{L}_{MC}(\theta) = \sum_{i=1}^{N} \frac{1}{m} \left\| A h_\theta(y_i) - y_i \right\|^2 \tag{6}$$

And the equivariance error function by:

$$\mathcal{L}_{EQ}(\theta) = \sum_{i=1}^{N} \sum_{g=1}^{|\mathcal{G}|} \frac{1}{|\mathcal{G}|\, n} \left\| T_g h_\theta(y_i) - h_\theta\left(A T_g h_\theta(y_i)\right) \right\|^2 \tag{7}$$

The global surrogate loss function we aim to minimize is:

$$\arg\min_\theta \sum_{i=1}^{N} \left(\frac{1}{m} \|A h_\theta(y_i) - y_i\|^2 + \alpha \sum_{g=1}^{|\mathcal{G}|} \frac{1}{|\mathcal{G}|n} \|T_g h_\theta(y_i) - h_\theta(A T_g h_\theta(y_i))\|^2 \right) \tag{8}$$

Where α is a positive trade-off parameter between the measurement consistency and equivariant loss.

This presents the advantage of not using any paired ground-truth data but only the measurements y_i and transformed and reconstructed version of them inside the loss. Experiments showed this loss function gave very good results with low noise and measurements degradation but suffers with high noise.

The Robust Equivariant Imaging (REI) loss proposed in [6], incorporating Steins Unbiased Risk Estimator (SURE) demonstrated strong reconstruction improvements at high noise levels, but suffered from disappointing results at lower noise levels. Both vanilla EI and REI assumed the noise level to be known, which does not fit the spirit of self-supervised learning, where the noise characteristics might not be known (very useful in real-world applications). Recently, the use of the UNSURE [8] tackled this

aspect. The unbiased and consistent estimator -under certain assumptions- provides a way to estimate the risk of a model without needing to know the noise level. Particularly, UNSURE demonstrated better performances at low noise level compared to SURE + EI in [8].

3. Dynamic Equivariant Imaging

Working on 2D-MRI sequences suggests we still have 2 spatial dimensions (defined by Height H and Width W), but we use the depth D as a temporal dimension T. Therefore, we work on a 2D spatial domain and on a 1D temporal domain. We interpret the axial dimension as time which is a trick, because in fMRI there is no anatomical structure's evolution over time, despite being the case with our dataset.

Similarly to simple 2D MRI, the ground-truth data is altered in the Fourier domain by subsampling some frequencies (Fourier coefficients). The difference is the unfixed aspect of the mask along the temporal axis. In 2D MRI, the same mask is applied over every images of the dataset. In Dynamic MRI, the same mask is applied over every volume of the dataset, but this mask is time-varying over its depth (time). Therefore each slice of the mask volume has a different subsampling.

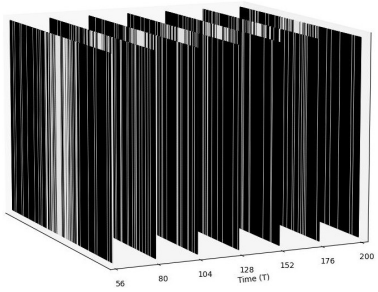


Figure 1. Time-Varying MRI Mask

3.1. Transforms

In [5] authors introduced projective projections. They have the potential to layer more information than often used rotations and/or flips as they combine multiple transforms.

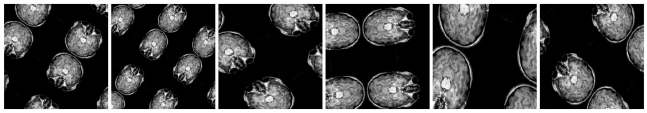


Figure 2. Homographic transforms [5]

As for now, the different groups of transformations have only been applied to spatial dimensions in the different research papers. As we innovate by working on dynamic data and not static data, we need to propose transformations satisfying the equivariance of $h \circ A$ in this dimension.

If in the spatial domain its intuitive to represent a 2D spatial transformation (e.g. rotation), this is non-sense in the temporal domain which is in one dimension only by definition. How could we rotate over the temporal axis ? Same logic goes with shearing... and other projective transformations that cant be applied to a temporal dimension. None of these are not applicable in the time domain, because its only in one dimension. However, symmetry transformation seems suited for 1D data .

This is the most evident transformation in the 1D temporal domain, with the flip of our measurement (i.e. sequence of images) alongside the T axis. If we reverse the direction of time, the sequence properties remain the same. The first frame of the sequence becomes the last of the symmetric one, and the last frame of the first sequence becomes the first one of the symmetric one. As the time axis is orthogonal and distinct from the two spatial axis, flipping the time does not impact the spatial transformations and matches the invariance property in temporal 1D.

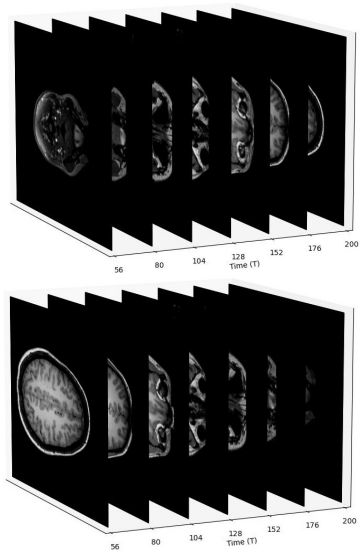


Figure 3. Transformed volume

4. Space-Time Modeling

4.1. Time handling for machine learning

In order to represent dynamic MRI data with Python, we can interpret a video (or other space + time format) as a batch of 2D images. However, this doesn't mean that we apply 2D denoising architectures. Dynamic MRI implies capturing a sequence of 2D image slices over time, therefore over a 3D piece of information. The additional dimension means we now work on voxels and not pixels, carrying

by definition more information. 3D denoisers can consequently better model and reduce noise by utilizing the additional information contained in this third dimension, leading to (an expected) improved image quality and more accurate reconstructions. Especially, consecutive frames often contain redundant information helping to smooth the sequence from one frame to another. We expect 3D denoisers to better leverage this redundancy to preserve dynamic structures and motion patterns in the MRI data, unlike 2D denoisers which treat each frame independently. The key advantage of 3D denoisers is their ability to utilize the full 3D structure to improve denoising and reconstruction.

The 3D denoisers do not make any distinction between a 3 spatial dimension volume or a 2D + time volume, this is why the use of 3D denoisers does not alter the physical processes underlying MRI and respects our initial objective. We refer to depth of 3D denoiser (H,W,D) but the time of a dynamic sequence (H,W,T).

4.2. Patching in dynamic MRI

3D unrolled models in high-resolution are the most consuming models we work on in this study. It is well known that unrolling is challenging for GPU usage. We can't extract patches in the Fourier space because in the frequency domain, the information about the exact spatial location of features is lost [9]. However, Fourier is time-independent (Fourier isn't related to time but only to space). As we work on dynamic MRI, we use a 3rd temporal dimension which is not considered as spatial. This is over this axis that we can patch the data to limit memory usage.

These patches depth "t" can be adjust to fit the GPU capacity, but we consider that under $t = 8$ slices in depth, it would not have enough information to capture the dynamic aspect. This suggest that with an initial batch of B volumes, we go from tensors of shape $(B, 2, T, H, W)$ to $(B\dot{(}T/t), 2, t, H, W)$.

We need to be careful with the acceleration mask we use so each patched mask of length t is different for a given volume, but is repeated with a modulo T so the same for each volume. Using these temporal patches, we recommend setting the train and test batch size to 1. We can interpret the patch as a replacement of the batch size, or a mini-batch. If you can set train batch size = 2, we recommend increasing t. In ideal scenario, $t = T$. In all the experiments we used a t that can divide T.

5. Experimental Setup

5.1. Data for dynamic MRI

The implementation of the DESSL reconstruction is based on the Queensland Twin IMaging (QTIM) dataset [10], which contains clinically-acquired anatomical and functional MRI scans. The QTIM dataset (available here) contains anatomical and functional scans of 1026 young adult twins and siblings aged between 18 and 30, as well as 176 twins aged 12 or 16. The T1-weighted (T1w) anatomical scans are already reconstructed with real and imaginary parts combined (only magnitude). We only work on the first anatomical scan of each participant, leading to a total database of 1202 NIfTI files . We use only 100 volumes for the experiment.

As there is a wide range of values for each pixel/voxel for the different patients, with few values reaching +450 and other going under 0, we apply a min-max scaling in function of the 99.5th percentile. Also, the volumes acquired in the QTIM dataset are not cleaned and lot of noise degrades the volumes quality. To deal with this issue, we denoise the database itself. To process a volume, it appears that a combination of TV-denoising followed by a ℓ_1-denoising is efficient. The TV-denoising removes the parasite noise inside the brain volume resulting in anisotropic diffusion and edge-preservation, while ℓ_1 denoising is good to remove the diffused redistributed noise by promoting sparsity in the denoising process. This second denoiser is desired to have flat "zero" areas around the volume. For reproducibility, we use:

Parameter	**Value**
Maximum Iterations	50
Step Size	0.5
Regularization Parameter (λ)	0.02
N Iter Max	25

Table 1. Parameters for TV-Denoising

Parameter	**Value**
Maximum Iterations	50
Step Size	0.25
Regularization Parameter (λ)	0.1

Table 2. Parameters for ℓ_1-Denoising

This leads to denoised volumes that are **almost** noise-free, meaning we can use them as trustful ground-truth to evaluate our models . The noise we add to the volumes to simulate MRI acquisition is not mixed with the original acquisition noise anymore.

A dynamic MRI scan is usually acquired over a single same frame (or volume) over time in clinic. As we do not possess such a database, we simulate dynamic MRI by flattening 3D volumes as a sequence of images. It consists

in using the depth (here time "T") dimension as the batch size of a volume. We then apply a time-varying mask over the flattened batch of images. From a 256x256x256 volume we now have for a single volume a tensor of shape (B*T,C,h,W) = (256,2,256,256). With the reverse operation we can rebuild the volume.

As we don't have relevant features over the firsts and lasts frames of most participants brains, we only work on frames 48 to 208 (128+/-80). This skips the empty sequence slices as well as the few features frames, while still working on various types of head slice. Applying a time-varying MRI Cartesian acceleration mask using Gaussian undersampling ($R = 4$ and $\sigma = 0.05$):

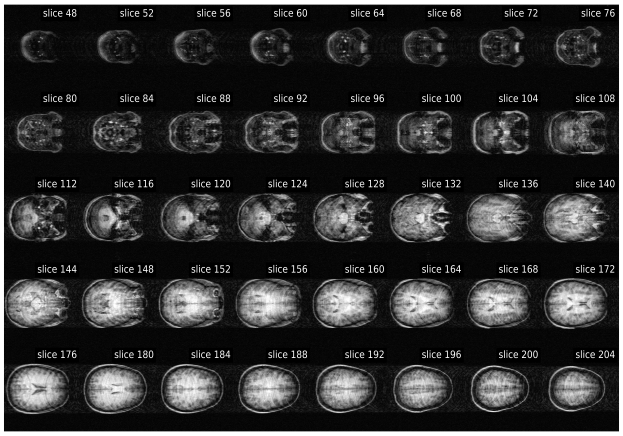


Figure 4. Flattened MRI Linear Reconstruction

5.2. Setup and Implementation

As there is the major transition to 3D, the memory usage in this experimentation is way bigger than usual 2D models. In order to train and test the different algorithms without expensive calculation time, we use CUDA (11.8 or 12.1) to use the GPU fast computation. However, to avoid saturating the GPU memory too fast, we will use diverse tricks (see [5.3] for details) to learn full-resolution models, and we will load and degrade the ground-truth dataset with the CPU to get the pairs of ground-truth / measurements.

For this project we use a simple NVIDIA RTX3050 laptop GPU with 4 GB of memory in a CUDA 12.1 environment for fixed parameters models, but a bigger station incorporating a RTX 3080 with 10 GB of memory in a CUDA 11.8 environment for unrolled dynamic models. This card has 8704 cuda cores allowing fast tensor computation.

The models are developed based on the Deep Inverse Library (Deep Inv) available here which is built on PyTorch. Small modifications have been applied to original library to pass the mini-batches in the dataloader.

5.3. Dynamic Unrolled Models

In this experiment, we set the patch depth to $t = 8$, meaning we work on smaller volumes of shape 256x256x8 instead of 256x256x160. Each volume then have 160/8 = 20 patches. For results reproducibility, we set for all unrolled models:

DnCNN config	**Value**
Batch Normalization	False
Bias	True
Depth	7
Number of Filters (nf)	32
Iterator Parameters	
iterator	ADMM
iter	5
Training Parameters	
Max Epochs	100
Learning Rate	5e-4

Table 3. Parameters for dynamic unrolled PnP-ADMM

Models were trained only once at 4 different noise levels ($\sigma = 0, 0.01, 0.05, 0.1$), and the checkpoint with the lowest EI loss (lowest loss for supervised) was kept for evaluation. More training are required to improve the accuracy of the proposed results. Decision was taken not to use the gradient in the EI loss (i.e. gradient does not propagate through T_g) due to experiments and as recommended for super-resolution in [11].

6. Results

6.1. Supervised Unrolled PnP-ADMM

As a point of comparison, we compare the different DESSL models with supervised models, supposed to achieve the best reconstructions. As we simulate dynamic MRI thanks to an -almost- fully sampled MRI volume, we must be aware that we would never be in possession of such data in real dynamic MRI scenario. We obtain:

σ	$A^\dagger$	Supervised
0.0	23.6±0.4	34.9±0.6
0.01	23.6±0.4	33.9±0.5
0.05	22.7±0.3	30.8±0.4
0.1	20.7±0.2	29.3±0.4

Table 4. Supervised Noise Comparison

The results are exceptional for every noise level. Remarkably, the complex PSNRs from the supervised models are consistently about 10 dB higher than those achieved with backprojection. The model is even able to correct the original acquisition artifacts!

6.2. Unsupervised Unrolled PnP-ADMM EI

We distinguish the no dynamic EI version (transforms only in spatial domain) from the "dyn. EI" version (with time flip). The two sub-types are compared to assess the relevance of the time flip. For both vanilla models we use the parameters:

EI Loss Parameters	
Transforms	Affine
Transforms per minibatch	1
No Grad	True
Weight EI loss (α)	1.0

Table 5. Parameters for EI loss

In the EI loss, every mini-batch is transformed according to a random affine transformation and this same transformation is applied on every slice of the small batch. This suggests that the transformations between two successive patches are different. In the dynamic EI version, in addition to the spatial transformation, an independent time flip can be applied over each small batch. We use a Bernoulli drawing with a probability of 50% for each small batch to apply the flip. We recall that this transformation is orthogonal to the spatial planes and therefore do not alter the spatial transformations. We obtain:

σ	$A^\dagger$	No Dyn. EI	Dyn. EI
0.0	23.6±0.4	32.2±0.6	33.1±0.6
0.01	23.6±0.4	31.5±0.6	32.0±0.5
0.05	22.7±0.3	26.5±0.2	26.6±0.2
0.1	20.7±0.2	21.7±0.1	21.9±0.1

Table 6. Unsupervised EI Noise Comparison

Results are similar between no dyn. model and dynamic EI but still we observe a non-negligible gain at $\sigma = 0$ and $\sigma = 0.01$ with respective improvements of 0.9 dB and 0.5 dB. The higher the noise is, the less relevant is the effect. On figures 5 and 6, the acceleration artifacts are very well corrected according to the error difference. Also, the small PSNR gain between no-dyn and dynamic versions is not human-eye visible and results in sequences extremely similar, this is why we only present results for dynamic version.

Regarding the relevance of the flip in the temporal domain, results are mitigated. If there is an obvious gain at small noise levels, this gain disappears as the noise increases. However it is now demonstrated that **incorporating this transform is preferable in vanilla EI** as it can only improve the reconstruction. Especially at low noise level, the EI loss was able to exploit this new transformation into knowledge for reconstruction.

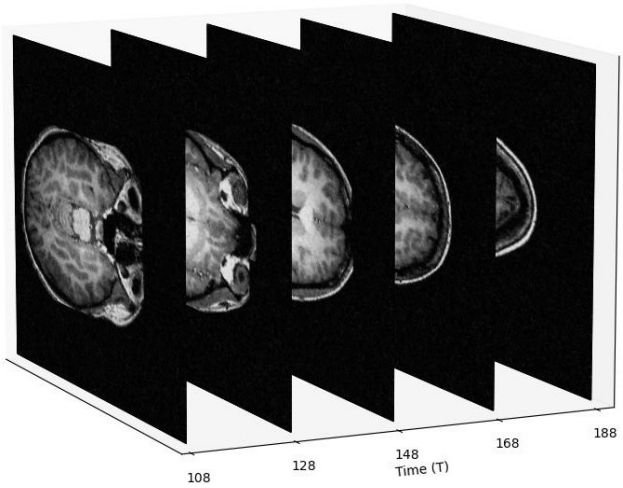


Figure 5. EI Flip Reconstruction

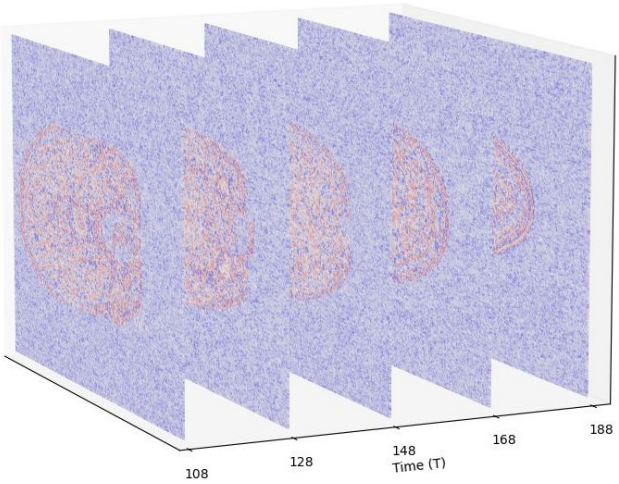


Figure 6. Error in EI Flip Model

6.3. Unsupervised Unrolled PnP-ADMM REI

The REI models only differ from the EI models by the use of the SURE loss to replace the MC loss and by the handling of noisy measurements in the equivariant loss.

REI Loss Parameters	
Transforms	Affine
Transforms per minibatch	1
No Grad	True
Weight EI loss (α)	1.0

Table 7. Parameters for REI loss

At noise level $\sigma = 0$, we use the results of EI, as REI = EI without noise by definition. This logic is suitable as in both vanilla EI and REI, we assume the noise distribution level

to be known. In [6], authors demonstrated the relevance of REI as soon as the noise level increased. We obtain after training:

σ	$A^\dagger$	No Dyn. REI	Dyn. REI
0.0	23.6±0.4	32.2±0.6	33.1±0.6
0.01	23.6±0.4	30.6±0.5	31.1±0.5
0.05	22.7±0.3	29.8±0.5	29.8±0.4
0.1	20.7±0.2	28.4±0.4	28.6±0.4

Table 8. Unsupervised REI Noise Comparison

Obtained results are outperforming EI models at strong noise level with a remarkable improvement of +7 dB for $\sigma = 0.1$, but is disappointing for σ = 0.01 with worse PSNRs. Regarding the differences between basic REI and dynamic REI, the results are once again better (at least not worse) but do not show an amazing improvement. The complex structures inside the brain are also accurately reconstructed, and no acceleration artifact is present. At sigma σ = 0.05, differences between the supervised model and no dyn. REI are not easy to catch. The error shows a very good denoising with residual noise in the central band (i.e. containing acceleration artifacts). The main visible difference with EI is the denoising around the brain. There is no residual noise and just some coefficients are badly estimated by the model but it's very insignificant and the model is very satisfying.

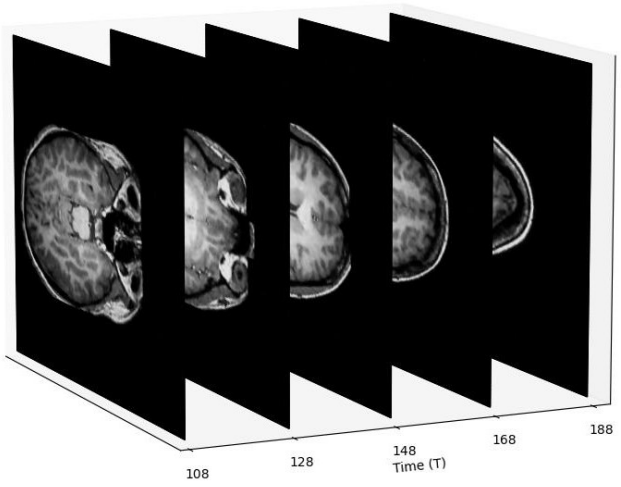


Figure 7. REI Flip Reconstruction

We can once again state that **incorporating the time transform is preferable in REI** and can only provide gain in the reconstruction. Possible reasons for the low gain as the noise increases are given in later sections. If the gain from EI to REI is not that big, the difference between linear reconstructions and REI models are impressive with always more than 7 dB of difference. Keeping in mind this

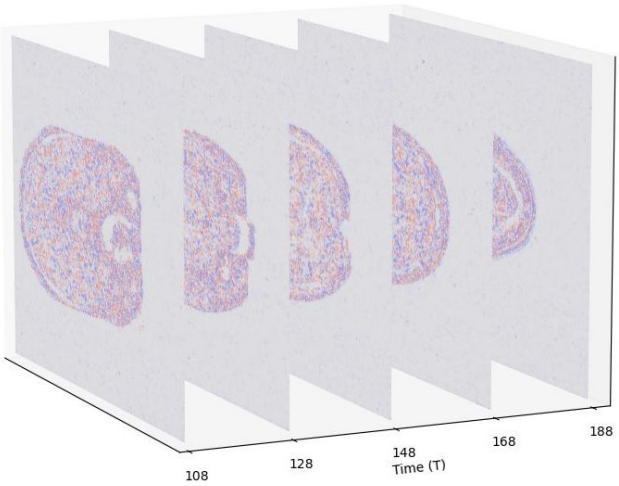


Figure 8. Error in REI Flip Model

model doesn't have access to the ground-truth data, the results are outstanding and flirt with what we obtain in supervised models.

6.4. Unsupervised Unrolled PnP-ADMM RUEI

Robust Unsure Equivariant Imaging (RUEI) uses the recently introduced UNSURE estimator in the loss term instead of the SURE loss term. The main difference is the unknown noise level for the SURE estimator which was usually assumed to be known. Computation parameters are the same as in table 7. The primary aim is to exceed the performance of REI at low noise levels after the disappointing results at $\sigma = 0.01$. We obtain:

σ	$A^\dagger$	No Dyn. RUEI	Dyn. RUEI
0.0	23.6±0.4	32.9±0.6	33.1±0.6
0.01	23.6±0.4	32.0±0.5	32.4±0.5
0.05	22.7±0.3	29.3±0.4	29.5±0.4
0.1	20.7±0.2	28.5±0.4	28.4±0.4

Table 9. Unsupervised RUEI Noise Comparison

The results brought by RUEI are extremely satisfying at every noise level and especially at $\sigma = 0.01$ where RUEI outperfoms REI of +1.3 dB for both dynamic and no dynamic versions. **RUEI demonstrates more stability and should be preferred over REI and vanilla EI**, with better regularity and matching with the spirit of inverse problems where noise is unknown by definition. The good performance of RUEI compared to REI at low noise level might enhances the importance of the pre-denoising step applied to the whole dataset, that might contain residual noise. Therefore, as REI assumes the noise distribution level to be known, giving a wrong σ can have repercussions at low noise. Visuals are close to the ones of REI models.

σ	$A^\dagger$	No Dyn. EI	Dyn. EI	No Dyn. REI	Dyn. REI	No Dyn. RUEI	Dyn. RUEI	Supervised
0.0	23.6±0.4	32.2±0.6	**33.1±0.6**	32.2±0.6	**33.1±0.6**	32.9±0.6	**33.1±0.6**	34.9±0.6
0.01	23.6±0.4	31.5±0.6	32.0±0.5	30.6±0.5	31.1±0.5	32.0±0.5	**32.4±0.5**	33.9±0.5
0.05	22.7±0.3	26.5±0.2	25.7±0.2	29.8±0.5	**29.8±0.4**	29.3±0.4	29.5±0.4	30.8±0.4
0.1	20.7±0.2	21.7±0.1	21.9±0.1	28.4±0.4	**28.6±0.4**	28.5±0.4	28.4±0.4	29.3±0.4

Table 10. PSNR Comparison

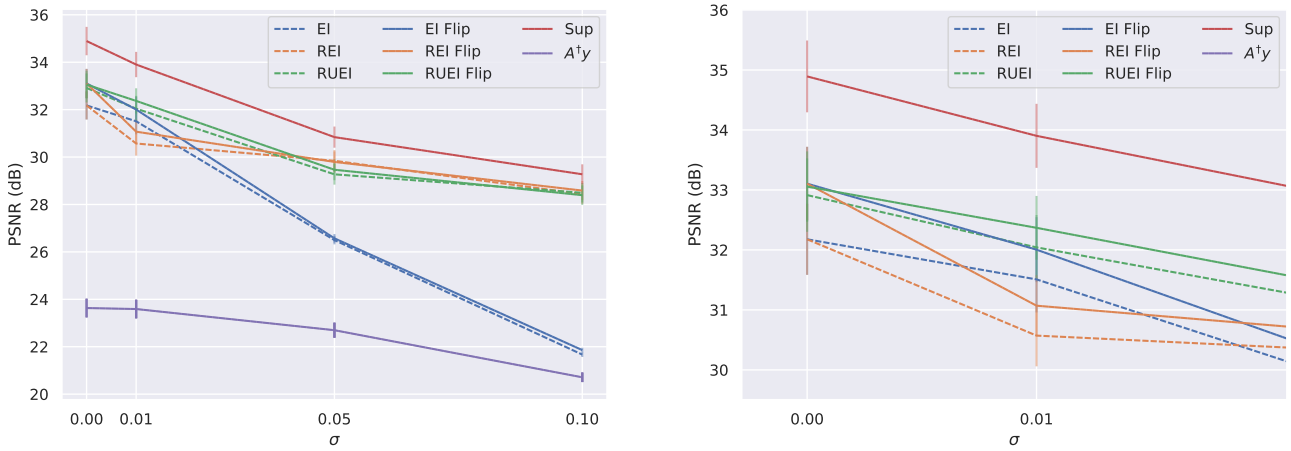


Figure 9. Unrolled Models Comparison for different σ

7. Discussions and Further Work

Table 10 and figure 9 highlight the behavior of the different DESSL models according to the noise level (if PSNR is equal, lowest std is preferred). If REI models yield the best result on 3 out of 4 noise level, RUEI is close behind and outperforms REI at low noise level. RUEI is more stable than any other EI model and its curves follows the trend of the supervised model one. The use of the dynamic transform is more effective at low noise levels than at high noise levels where its relevance can be discussed. Nevertheless, dynamic models have better or equals results compared to no dynamic models in 11 out of 12 comparisons (with no dyn. RUEI better of 0.1 dB at $\sigma = 0.1$).

Several phenomena were observed and must be highlighted:

- The SURE losses becomes negative very quickly. It is very surprising to always obtain a negative SURE (and UNSURE). We can't pronounce ourselves if it had a big impact on the reconstruction or not as we were not able to keep it positive. Still, even when the term was inferior to 0, it kept decreasing progressively.
- The EI loss term can fluctuate significantly from one epoch to another. It is not a good thing and shows a lack of stability. This can be linked to the absence of gradient propagation in the EI loss. An attempt using "no grad" to False at $\sigma = 0.05$ showed more stability in the EI loss and therefore it might be interesting to continue the researches in that way.
- A single attempt using REI at noise level 0.01, but with a σ of 0.007 in the SURE loss, showed an increase of 0.5 dB (31.7 dB) compared to $\sigma = 0.01$ but is still far from the vanilla EI result. It finished with divergence. This joins the idea of residual noise in the -considered as- ground truth denoised dataset that can bias the SURE loss with assumed known noise. RUEI does not suffer from such an issue as it estimates the noise level by itself.
- Using a batch of one is very noisy and the training is sensitive to the last iteration of an epoch, influencing a lot the final model.
- Using only one transform per mini-batch was sufficient to prove EI, REI and RUEI skills and increasing the number of transformations just increased the computation time.

8. Conclusion

Is dynamic EI able to capture very fine structures in brain MRI?
Reconstructions showed impressive results and this for every noise level. Particularly at high noise levels, REI incorporating the dynamic transform demonstrated the relevance of equivariant imaging to compete with supervised models which is a massive achievement. Therefore the answer is yes.

How does the EI loss deal with a new dimension?
This question is more nuanced. We obtained results that were not very stable possibly due the absence of gradient in the EI loss in the conditions of our experiment. Therefore if the EI loss was easy to adapt for dynamic data and still achieved impressive results, the optimal parameters for dynamic MRI might not have been found yet.

Which transformations can we apply in dynamic MRI?
The temporal symmetry over the time dimension showed improvements particularly at low noise level. We recommend keeping this transformation. Regarding the proposed group transformation with the different affine transformations that evolves over time, we can't conclude as brain MRI might not be the best scenario to test it, and we didn't implement it. As a general statement, **incorporating the time transform is preferable in dynamic self-supervised learning with equivariant imaging**. Even if we do not have a big gain at high noise level, it always showed improvements, either in EI, REI or RUEI. This reduced impact with increasing noise level can be attributed to the robust nature of the robust models, which, due to its use of the SURE loss for consistency and noise application in the EI loss, diminishes the importance of temporal transformations.

Is EI sensitive to a limited number of transformations over the temporal axis (with comparison to a combination of transformations in spatial 3D) ?
We can only partially conclude. To the question "is it worthy to use only one transformation compared to zero", the answer is yes. The time flip showed a relative gain of knowledge and therefore demonstrated that even with only one transformation, we can gain more information (compared to 0). However, we cannot conclude to the question "would we gain more information with a group of spatial transformations" because of the difficulty to build a full 3 spatial dimensions model due to hardware limitations.

What version of Equivariant Imaging is preferable in dynamic MRI ?
The difference between EI and robust models becomes more pronounced at higher noise levels. EI struggles significantly with noise levels above $\sigma = 0.01$, while robust models continue to perform well even under high noise conditions, demonstrating their robustness. **RUEI demonstrated regularity for every noise level and should be preferred**. This is also more rigorous regarding ill-posed inverse problems nature with unknown parameters.

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